

Task 5: Exploratory Data Analysis (EDA)

Titanic Dataset Analysis

Internship: Data Analyst Internship

Task: Exploratory Data Analysis

Tools Used: Python, Pandas, Matplotlib, Seaborn

Objective

The objective of this task is to perform Exploratory Data Analysis (EDA) on the Titanic dataset to identify important patterns, trends, relationships, missing values, and outliers using statistical and visual analysis.

Introduction

Exploratory Data Analysis (EDA) is an important step in data analysis that helps understand the structure and characteristics of a dataset.

In this task, the Titanic dataset is analyzed using Python libraries such as Pandas, Matplotlib, and Seaborn. The analysis focuses on passenger demographics, survival patterns, passenger class, age, fare, missing values, relationships between variables, and potential outliers.

Various statistical methods and visualizations such as count plots, histograms, boxplots, scatterplots, heatmaps, and pairplots are used to extract meaningful insights from the data.

EDA Methodology

The following steps were performed during the analysis:

1. Loaded the Titanic dataset using Seaborn.
2. Inspected the dataset structure, shape, columns, and data types.
3. Checked descriptive statistics using `describe()`.
4. Identified missing values in the dataset.
5. Analyzed categorical variables such as gender, survival status, passenger class, and embarkation port.
6. Used visualizations to study distributions and relationships between variables.
7. Analyzed age and fare distributions using histograms and boxplots.
8. Examined relationships using scatterplots, correlation heatmaps, and pairplots.
9. Calculated survival rates based on gender, passenger class, and age groups.
10. Performed multivariate analysis to understand the combined effect of gender, passenger class, and survival.

11. Summarized the major findings and insights obtained from the analysis.

Dataset Overview

The Titanic dataset contains information about passengers who travelled on the Titanic.

The dataset includes demographic, passenger class, family, fare, and survival-related information. It contains both numerical and categorical variables, making it suitable for exploratory data analysis.

Important Variables

- **Survived:** Indicates whether the passenger survived (0 = No, 1 = Yes).
- **Pclass:** Passenger class (1 = First, 2 = Second, 3 = Third).
- **Gender:** Passenger gender.
- **Age:** Age of the passenger.
- **SibSp:** Number of siblings or spouses aboard.
- **Parch:** Number of parents or children aboard.
- **Fare:** Passenger ticket fare.
- **Embarked:** Port from which the passenger embarked.

The analysis uses statistical summaries and different visualizations to understand patterns and relationships among these variables.

Statistical Analysis

Statistical analysis was performed to understand the basic characteristics of the Titanic dataset.

The following methods were used:

- `df.shape` to identify the number of records and variables.
- `df.info()` to examine column names, data types, and non-null values.
- `df.describe()` to obtain statistical summaries of numerical variables.
- `value_counts()` to analyze the frequency of categorical values.
- `isnull().sum()` to identify missing values.
- `groupby()` to calculate survival rates for different passenger groups.
- `crosstab()` to compare relationships between categorical variables.

These methods provided a basic understanding of the dataset before performing detailed visual analysis.

Data Visualization

Data visualization was used to understand patterns and relationships in the Titanic dataset.

Different types of charts were created during the analysis, including:

- Count plots for categorical variables.
- Histograms for numerical distributions.
- Boxplots for identifying spread and potential outliers.
- Scatterplots for examining relationships between numerical variables.
- Bar plots for comparing survival rates.
- Correlation heatmaps for understanding relationships between numerical variables.
- Pairplots for multivariate analysis.

Each visualization was followed by observations to identify meaningful patterns and insights from the data.

Conclusion

The Exploratory Data Analysis of the Titanic dataset revealed several important patterns and relationships.

The analysis showed that survival was strongly associated with gender and passenger class. Female passengers generally had higher survival rates than male passengers, while First Class passengers had better survival outcomes compared with Third Class passengers.

The analysis also identified missing values, variations in age and fare distributions, and potential outliers. Different visualizations helped in understanding the relationships between passenger characteristics and survival.

Overall, EDA provided useful insights into the Titanic dataset and demonstrated how statistical analysis and visualization can be used to discover meaningful patterns in real-world data.

```
In [1]: # import the required library
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load Titanic dataset
df = sns.load_dataset("titanic")

# Display first 5 rows
df.head()
```

Out[1]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adul
0	0	3	male	22.0	1	0	7.2500	S	Third	man	
1	1	1	female	38.0	1	0	71.2833	C	First	woman	
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	
3	1	1	female	35.0	1	0	53.1000	S	First	woman	
4	0	3	male	35.0	0	0	8.0500	S	Third	man	

In [2]:

```
# Check the shape of the dataset
print("Number of rows and columns:", df.shape)

# Display column names
print("\nColumn Names:")
print(df.columns)

# Display dataset information
print("\nDataset Information:")
df.info()
```

Number of rows and columns: (891, 15)

Column Names:

```
Index(['survived', 'pclass', 'sex', 'age', 'sibsp', 'parch', 'fare',
      'embarked', 'class', 'who', 'adult_male', 'deck', 'embark_town',
      'alive', 'alone'],
      dtype='object')
```

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 891 entries, 0 to 890

Data columns (total 15 columns):

#	Column	Non-Null Count	Dtype
0	survived	891 non-null	int64
1	pclass	891 non-null	int64
2	sex	891 non-null	object
3	age	714 non-null	float64
4	sibsp	891 non-null	int64
5	parch	891 non-null	int64
6	fare	891 non-null	float64
7	embarked	889 non-null	object
8	class	891 non-null	category
9	who	891 non-null	object
10	adult_male	891 non-null	bool
11	deck	203 non-null	category
12	embark_town	889 non-null	object
13	alive	891 non-null	object
14	alone	891 non-null	bool

dtypes: bool(2), category(2), float64(2), int64(4), object(5)

memory usage: 80.7+ KB

In [3]:

```
# Statistical summary of numerical columns
df.describe()
```

Out[3]:

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
In [4]: # Check missing values in each column
missing_values = df.isnull().sum()

print("Missing Values in Each Column:")
print(missing_values)
```

Missing Values in Each Column:

```
survived      0
pclass        0
sex           0
age          177
sibsp         0
parch         0
fare          0
embarked      2
class         0
who           0
adult_male    0
deck        688
embark_town   2
alive         0
alone         0
dtype: int64
```

```
In [5]: # Display only columns with missing values
print("Columns with Missing Values:")
print(missing_values[missing_values > 0])
```

Columns with Missing Values:

```
age          177
embarked      2
deck        688
embark_town   2
dtype: int64
```

Observation:

The Titanic dataset contains missing values in the Age, Embarked, Deck, and Embark_town columns. The Deck column has the highest number of missing values, while Embarked and Embark_town have only a few missing values.

```
In [6]: # Count passengers by gender
gender_counts = df["sex"].value_counts()

print("Passenger Count by Gender:")
print(gender_counts)
```

```
Passenger Count by Gender:
sex
male      577
female    314
Name: count, dtype: int64
```

Observation:

The dataset contains both male and female passengers. The `value_counts()` function helps us understand the distribution of passengers by gender.

```
In [7]: # Count survived and non-survived passengers
survival_counts = df["survived"].value_counts()

print("Survival Count:")
print(survival_counts)
```

```
Survival Count:
survived
0      549
1      342
Name: count, dtype: int64
```

Observation:

The `survived` column shows the number of passengers who survived and did not survive. A value of 0 represents passengers who did not survive, while 1 represents passengers who survived.

```
In [8]: # Rename sex column to gender
df.rename(columns={"sex": "gender"}, inplace=True)

# Check column names
print(df.columns)
```

```
Index(['survived', 'pclass', 'gender', 'age', 'sibsp', 'parch', 'fare',
       'embarked', 'class', 'who', 'adult_male', 'deck', 'embark_town',
       'alive', 'alone'],
      dtype='object')
```

```
In [9]: # Survival count by gender
survival_gender = pd.crosstab(df["gender"], df["survived"])

print("Survival by Gender:")
print(survival_gender)
```

```
Survival by Gender:
survived  0    1
gender
female    81  233
male     468  109
```

```
In [10]: # Survival count by passenger class
survival_class = pd.crosstab(df["pclass"], df["survived"])

print("Survival by Passenger Class:")
print(survival_class)
```

Survival by Passenger Class:

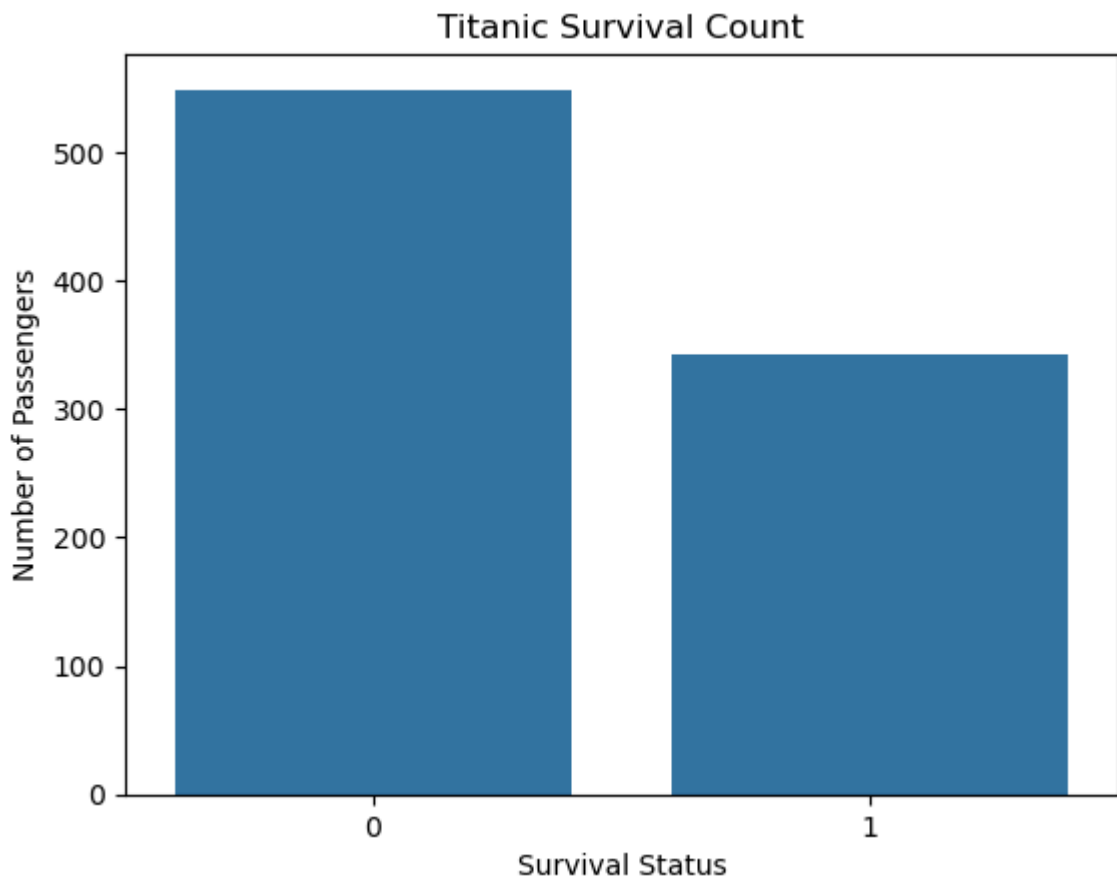
survived	0	1
pclass		
1	80	136
2	97	87
3	372	119

```
In [11]: # Survival Count Visualization

sns.countplot(data=df, x="survived")

plt.title("Titanic Survival Count")
plt.xlabel("Survival Status")
plt.ylabel("Number of Passengers")

plt.show()
```



Observation:

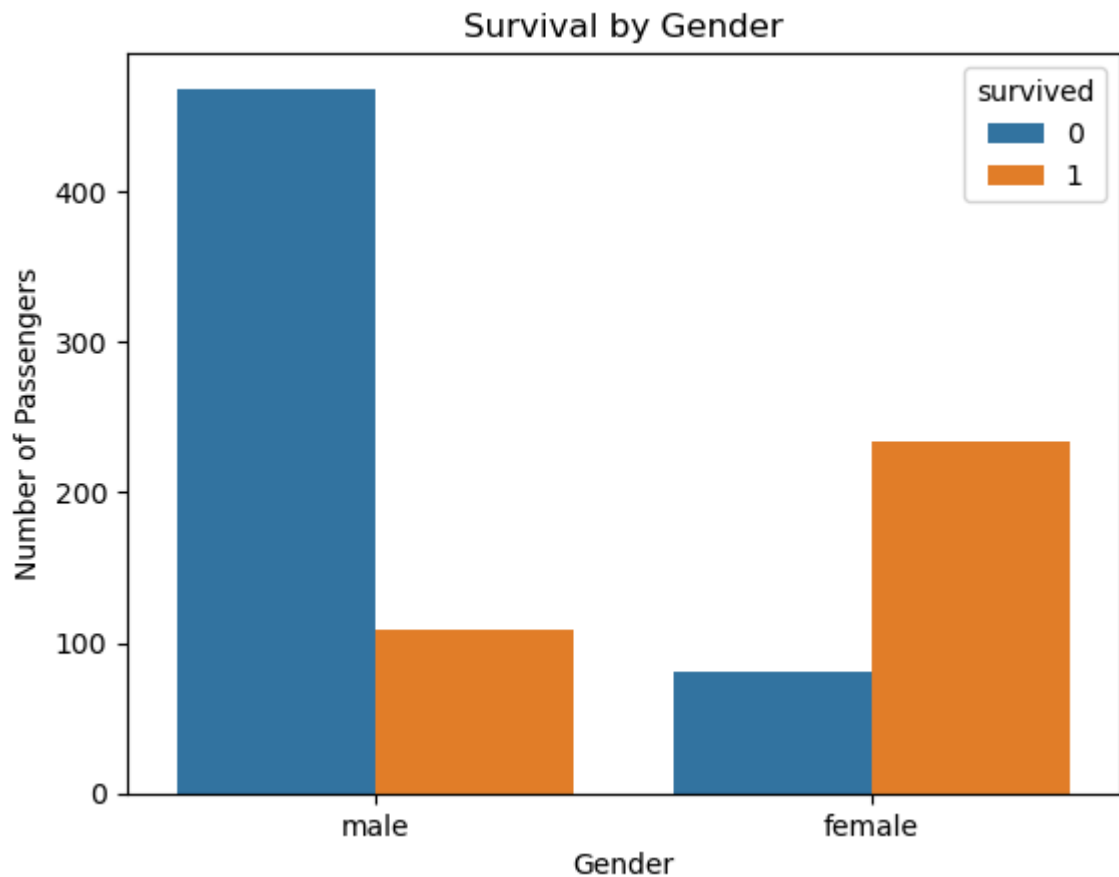
The bar chart shows the number of passengers who survived and did not survive. The number of passengers who did not survive is higher than the number of passengers who survived.

```
In [12]: # Survival by Gender Visualization

sns.countplot(data=df, x="gender", hue="survived")

plt.title("Survival by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")

plt.show()
```



Observation:

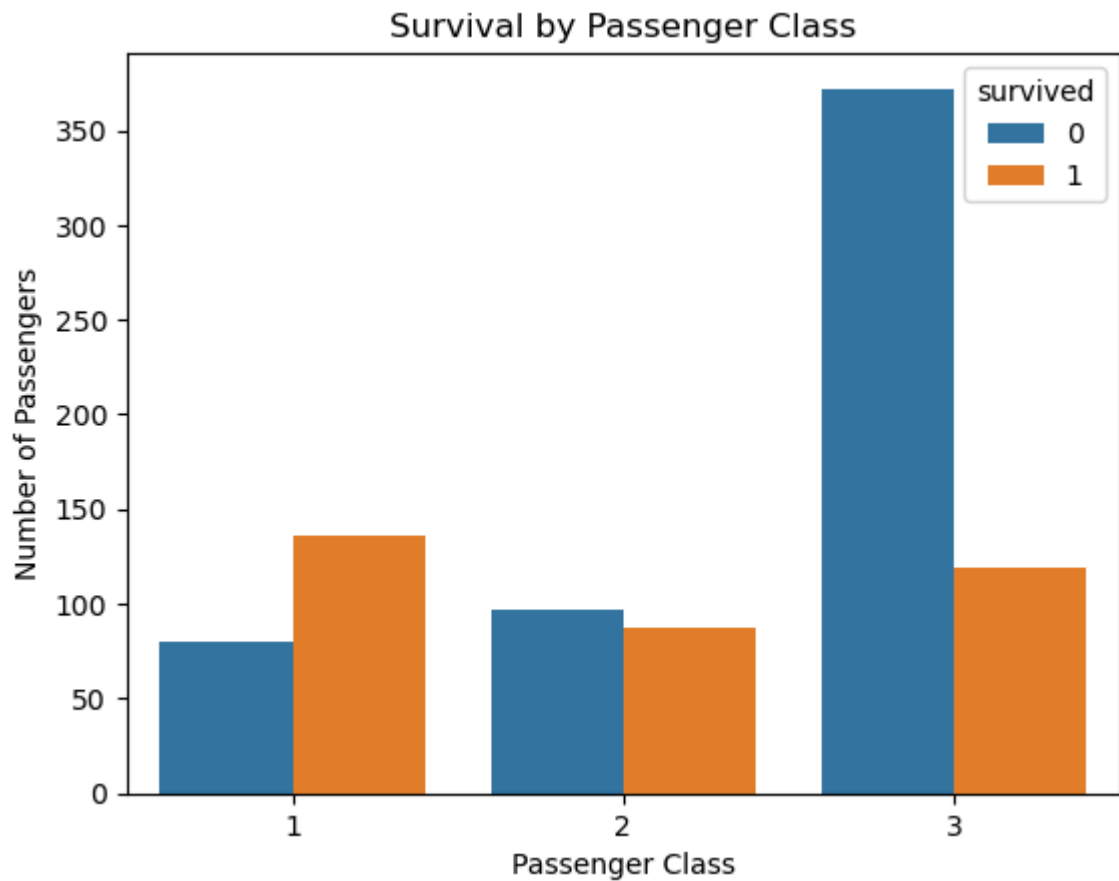
The chart compares survival outcomes between male and female passengers. It shows that survival outcomes differed by gender, with female passengers having a higher survival count relative to their total number of passengers.

```
In [13]: # Survival by Passenger Class Visualization

sns.countplot(data=df, x="pclass", hue="survived")

plt.title("Survival by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")

plt.show()
```

Observation:

The chart compares survival outcomes across the three passenger classes. It helps identify differences in survival between First, Second, and Third Class passengers.

 Chart meaning

1 = First Class

2 = Second Class

3 = Third Class

survived = 0 → Did not survive

survived = 1 → Survived

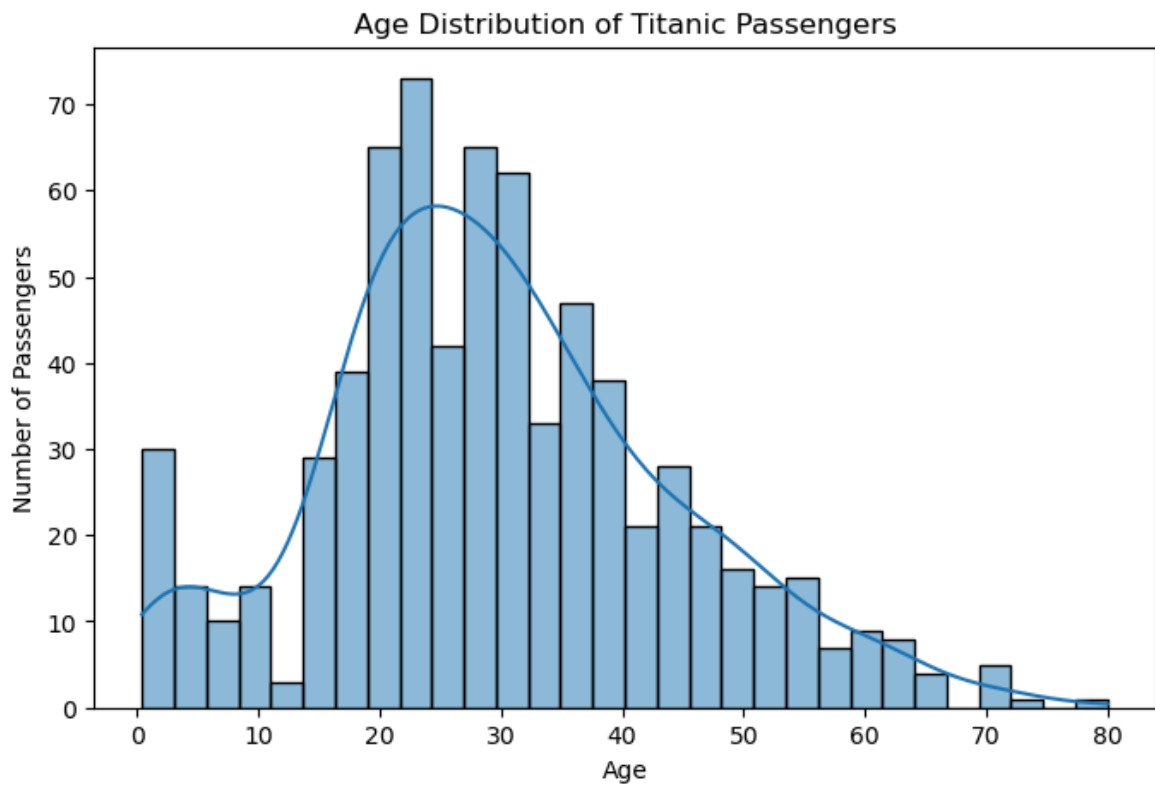
```
In [14]: # Age Distribution

plt.figure(figsize=(8, 5))

sns.histplot(data=df, x="age", bins=30, kde=True)

plt.title("Age Distribution of Titanic Passengers")
plt.xlabel("Age")
plt.ylabel("Number of Passengers")

plt.show()
```



Observation:

The histogram shows the distribution of passenger ages in the Titanic dataset. Most passengers were concentrated in the young and middle-age groups, while relatively fewer passengers were present at very young and very old ages.

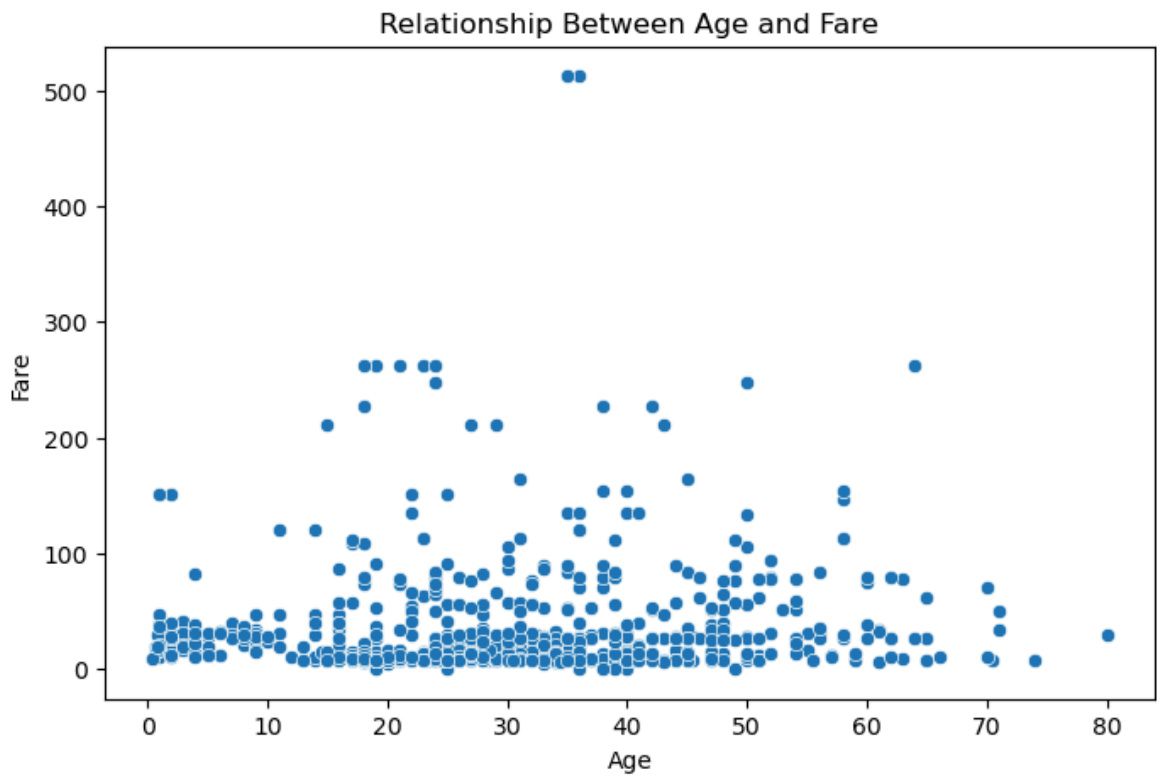
```
In [15]: # Age vs Fare Scatter Plot

plt.figure(figsize=(8, 5))

sns.scatterplot(data=df, x="age", y="fare")

plt.title("Relationship Between Age and Fare")
plt.xlabel("Age")
plt.ylabel("Fare")

plt.show()
```



Observation:

The scatter plot shows the relationship between passenger age and fare. The points are widely distributed, indicating that there is no strong clear relationship between age and fare. Some passengers paid unusually high fares compared with most passengers.

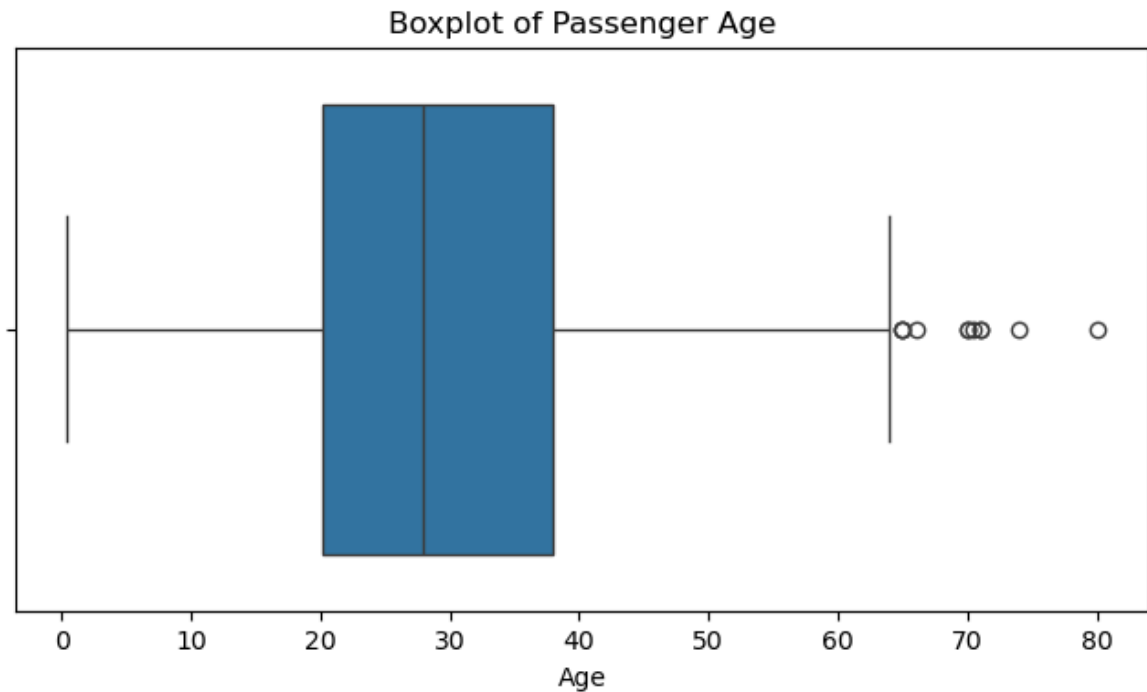
```
In [16]: # Age Boxplot

plt.figure(figsize=(8, 4))

sns.boxplot(data=df, x="age")

plt.title("Boxplot of Passenger Age")
plt.xlabel("Age")

plt.show()
```



Observation:

The boxplot shows the distribution of passenger ages and helps identify the median, spread, and possible outliers. A few points may appear outside the whiskers, representing unusually high or low ages.

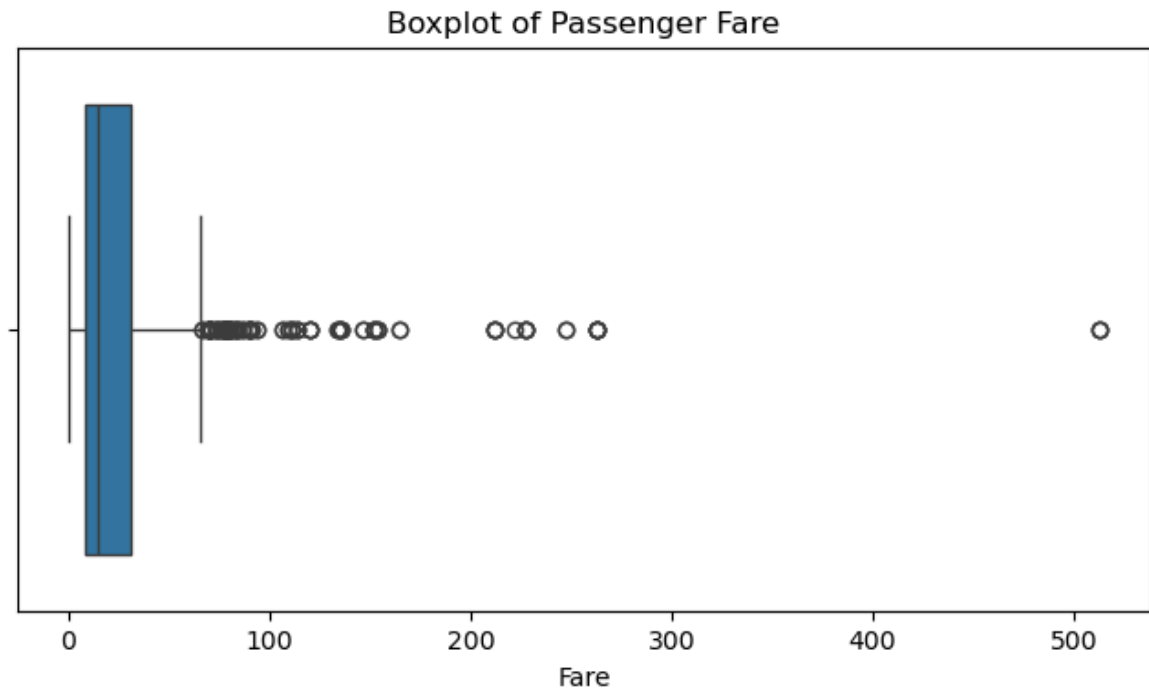
```
In [17]: # Fare Boxplot

plt.figure(figsize=(8, 4))

sns.boxplot(data=df, x="fare")

plt.title("Boxplot of Passenger Fare")
plt.xlabel("Fare")

plt.show()
```



Observation:

The boxplot shows the distribution of passenger fares. Most passengers paid relatively lower fares, while a small number of passengers paid much higher fares, which appear as potential outliers.

```
In [18]: # Correlation Heatmap

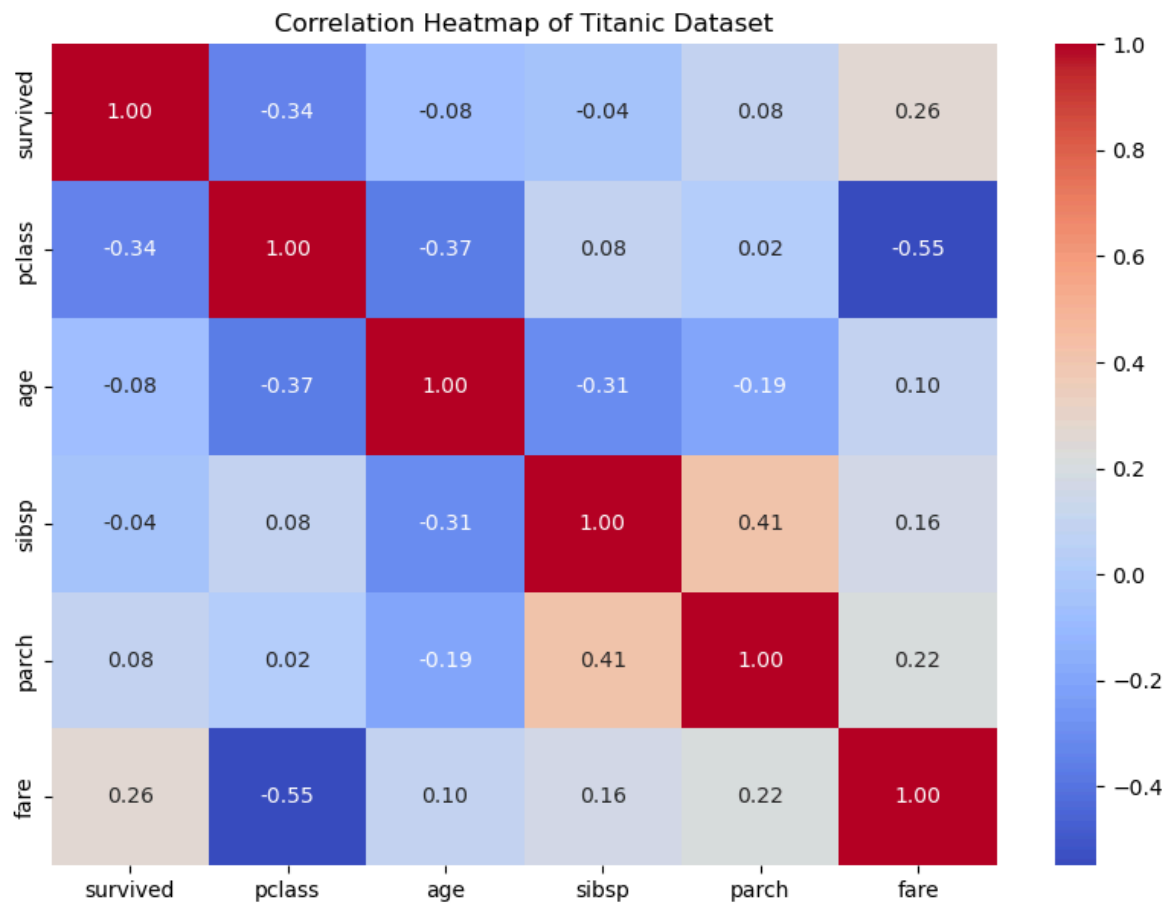
plt.figure(figsize=(10, 7))

correlation = df.select_dtypes(include="number").corr()

sns.heatmap(correlation, annot=True, cmap="coolwarm", fmt=".2f")

plt.title("Correlation Heatmap of Titanic Dataset")

plt.show()
```



Observation:

The correlation heatmap shows the relationships between numerical variables in the Titanic dataset. It helps identify variables that have positive or negative relationships with each other and with passenger survival.

```
In [19]: # Pairplot of Titanic Dataset

sns.pairplot(
    df[["survived", "pclass", "age", "sibsp", "parch", "fare"]].dropna(),
    hue="survived"
)

plt.suptitle("Pairplot of Titanic Dataset", y=1.02)

plt.show()
```



Observation:

The pairplot provides a visual overview of relationships among the numerical variables. It helps identify patterns, distributions, and possible relationships between passenger characteristics and survival.

```
In [20]: # Calculate survival rate by gender

gender_survival_rate = df.groupby("gender")["survived"].mean() * 100

print("Survival Rate by Gender:")
print(gender_survival_rate)
```

```
Survival Rate by Gender:
gender
female    74.203822
male      18.890815
Name: survived, dtype: float64
```

Observation:

The survival rate differs between male and female passengers. The analysis helps identify the relationship between gender and the likelihood of survival.

```
In [21]: # Calculate survival rate by passenger class

class_survival_rate = df.groupby("pclass")["survived"].mean() * 100

print("Survival Rate by Passenger Class:")
print(class_survival_rate)
```

Survival Rate by Passenger Class:

pclass

1 62.962963

2 47.282609

3 24.236253

Name: survived, dtype: float64

Observation:

The survival rate varies across passenger classes. Passengers in higher classes generally had a higher survival rate than passengers in lower classes. This indicates that passenger class was an important factor associated with survival.

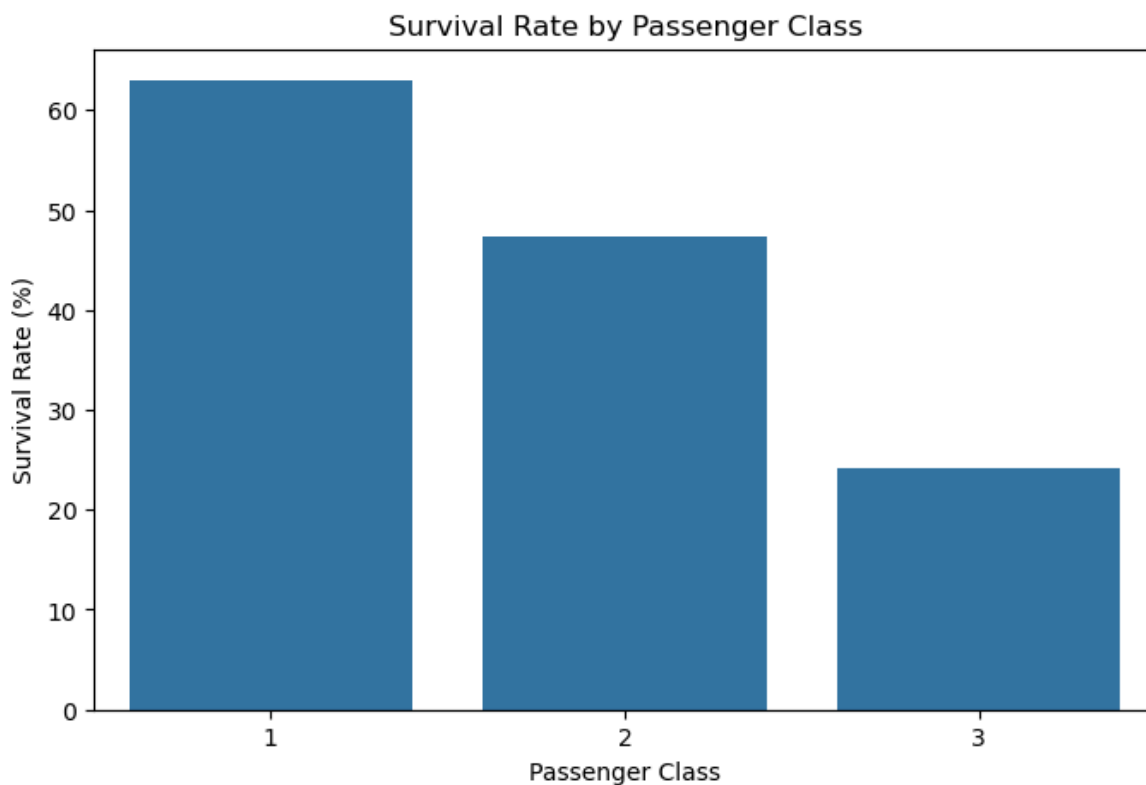
```
In [22]: # Survival Rate by Passenger Class

plt.figure(figsize=(8, 5))

sns.barplot(
    x=class_survival_rate.index,
    y=class_survival_rate.values
)

plt.title("Survival Rate by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Survival Rate (%)")

plt.show()
```

Observation:

The bar chart shows that survival rates were different across passenger classes. First Class passengers had the highest survival rate, while Third Class passengers had the lowest survival rate. This suggests that passenger class was strongly associated with survival.

```
In [23]: # Gender Distribution by Passenger Class

gender_class = pd.crosstab(df["pclass"], df["gender"])

print("Gender Distribution by Passenger Class:")
print(gender_class)
```

Gender Distribution by Passenger Class:

gender	female	male
pclass		
1	94	122
2	76	108
3	144	347

Observation:

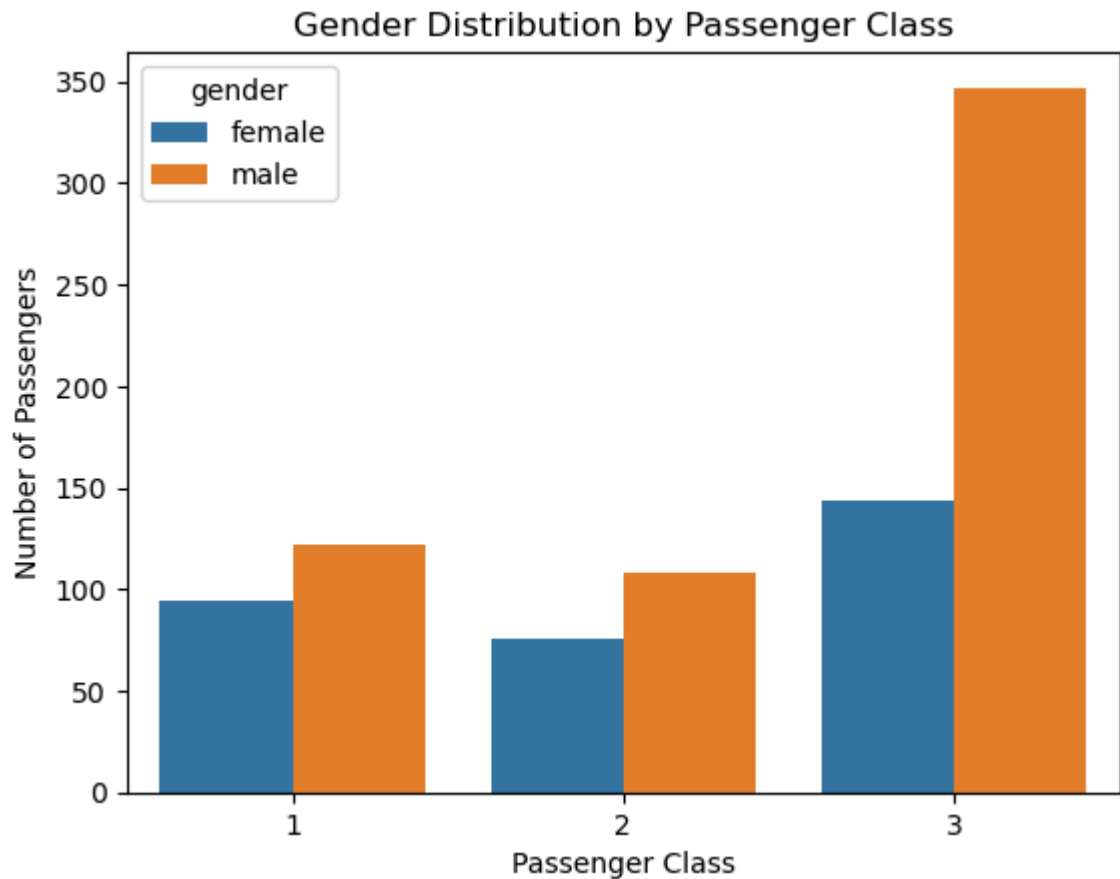
The table shows the distribution of male and female passengers across the three passenger classes. It helps us understand how gender was distributed among First, Second, and Third Class passengers.

```
In [24]: # Gender Distribution by Passenger Class Visualization

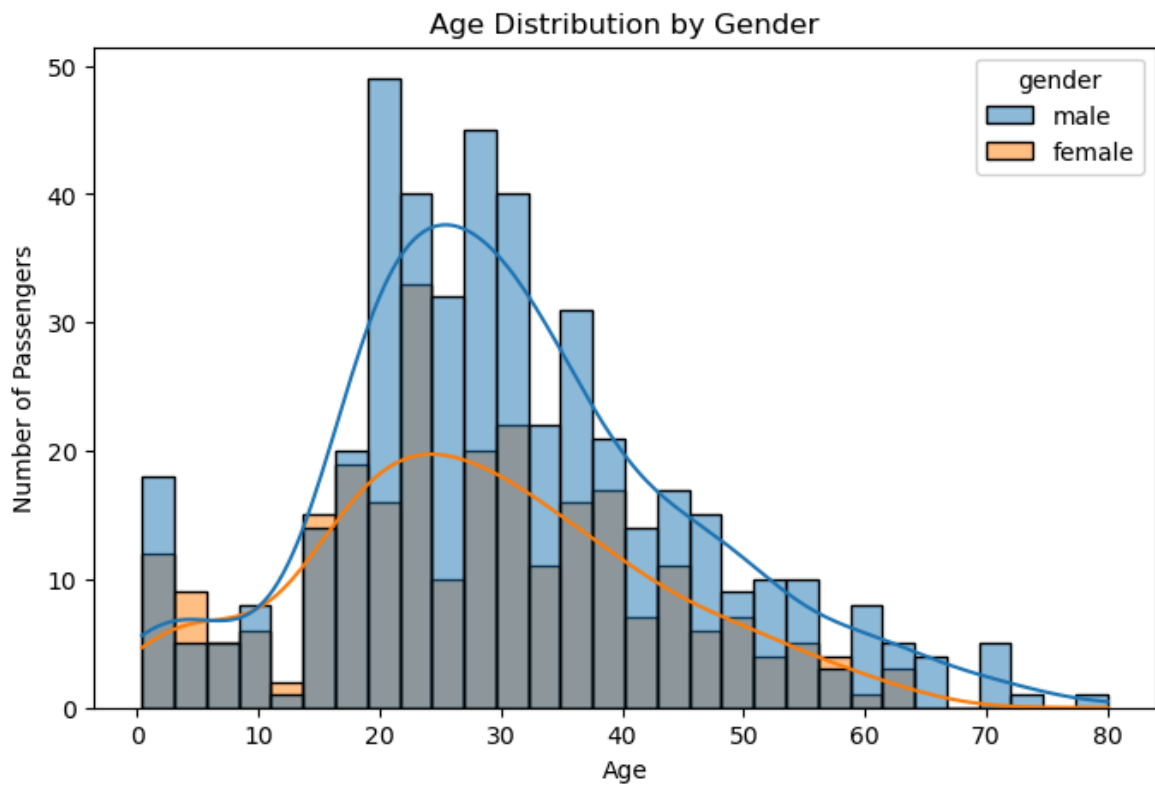
sns.countplot(data=df, x="pclass", hue="gender")

plt.title("Gender Distribution by Passenger Class")
plt.xlabel("Passenger Class")
```

```
plt.ylabel("Number of Passengers")  
plt.show()
```



```
In [25]: # Age Distribution by Gender  
  
plt.figure(figsize=(8, 5))  
  
sns.histplot(  
    data=df,  
    x="age",  
    hue="gender",  
    bins=30,  
    kde=True  
)  
  
plt.title("Age Distribution by Gender")  
plt.xlabel("Age")  
plt.ylabel("Number of Passengers")  
  
plt.show()
```



Observation:

The histogram compares the age distribution of male and female passengers. It shows how the ages of passengers were distributed across the two genders and helps identify the most common age groups.

```
In [26]: # Check for duplicate records

duplicate_count = df.duplicated().sum()

print("Number of Duplicate Rows:", duplicate_count)
```

Number of Duplicate Rows: 107

Observation:

The duplicate check helps identify whether the dataset contains repeated records. Duplicate records can affect the accuracy of exploratory analysis and should be considered before further analysis.

```
In [27]: # Check data types of all columns

print("Data Types of Each Column:")
print(df.dtypes)
```

Data Types of Each Column:

survived	int64
pclass	int64
gender	object
age	float64
sibsp	int64
parch	int64
fare	float64
embarked	object
class	category
who	object
adult_male	bool
deck	category
embark_town	object
alive	object
alone	bool

dtype: object

Observation:

The dataset contains both numerical and categorical variables. Numerical columns include variables such as age, fare, pclass, and survived, while categorical columns include gender, embarked, class, and other descriptive attributes.

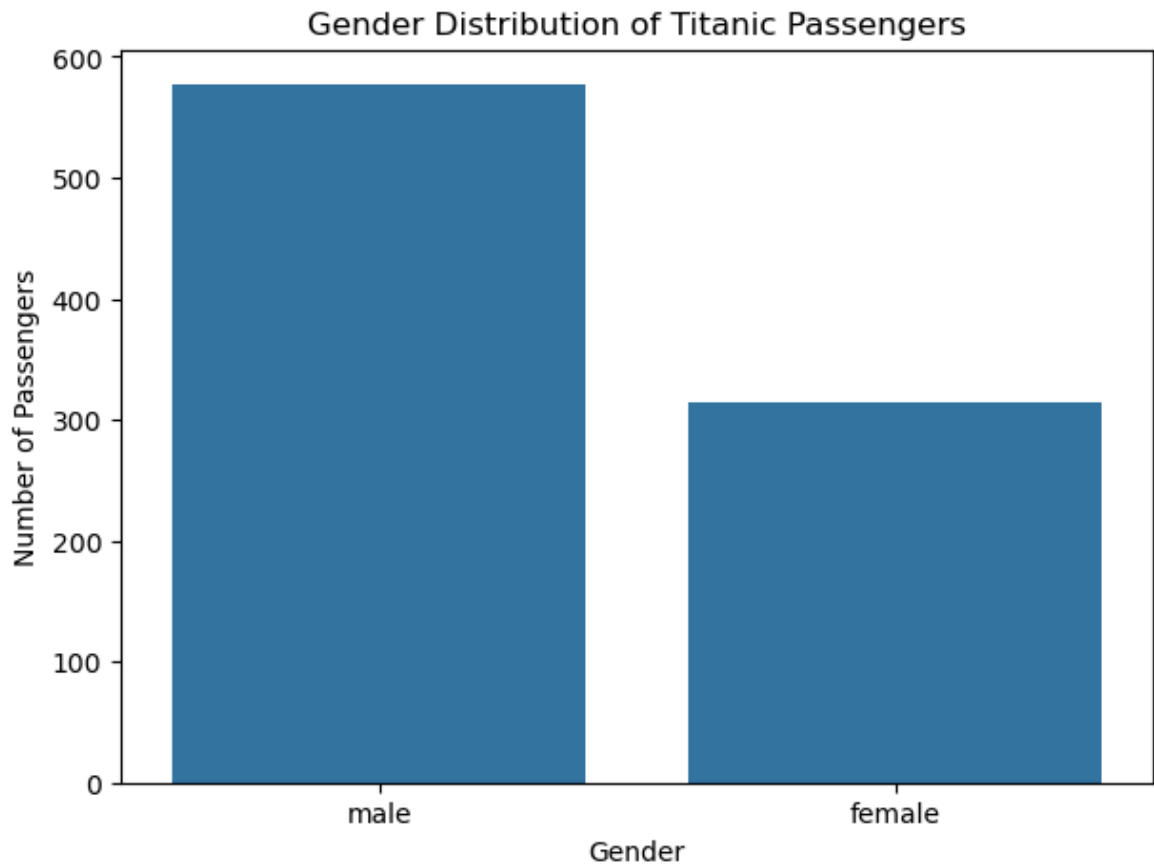
```
In [28]: # Gender Distribution

plt.figure(figsize=(7, 5))

sns.countplot(data=df, x="gender")

plt.title("Gender Distribution of Titanic Passengers")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")

plt.show()
```



Observation:

The bar chart shows the distribution of male and female passengers in the Titanic dataset. It helps us understand the number of passengers belonging to each gender category.

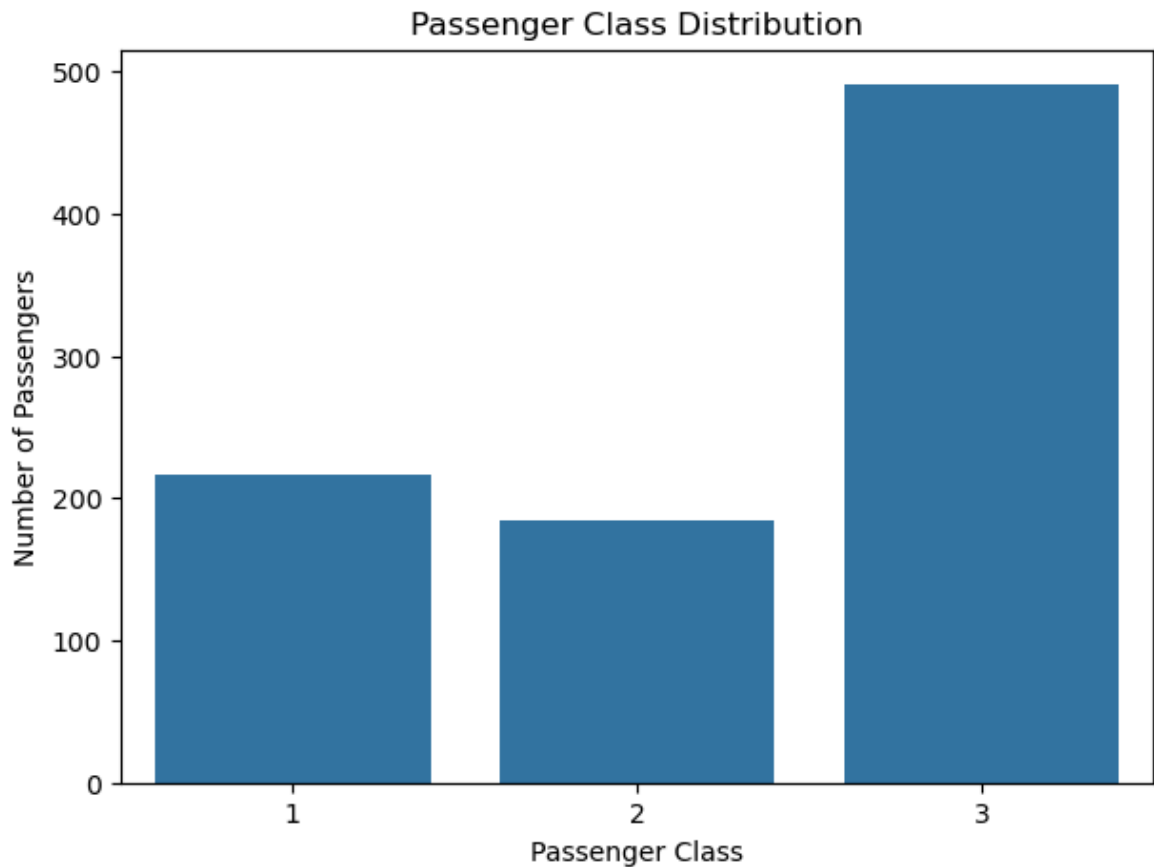
```
In [29]: # Passenger Class Distribution

plt.figure(figsize=(7, 5))

sns.countplot(data=df, x="pclass")

plt.title("Passenger Class Distribution")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")

plt.show()
```



Observation:

The bar chart shows the distribution of passengers across the three passenger classes. It helps identify which passenger class had the highest and lowest number of passengers.

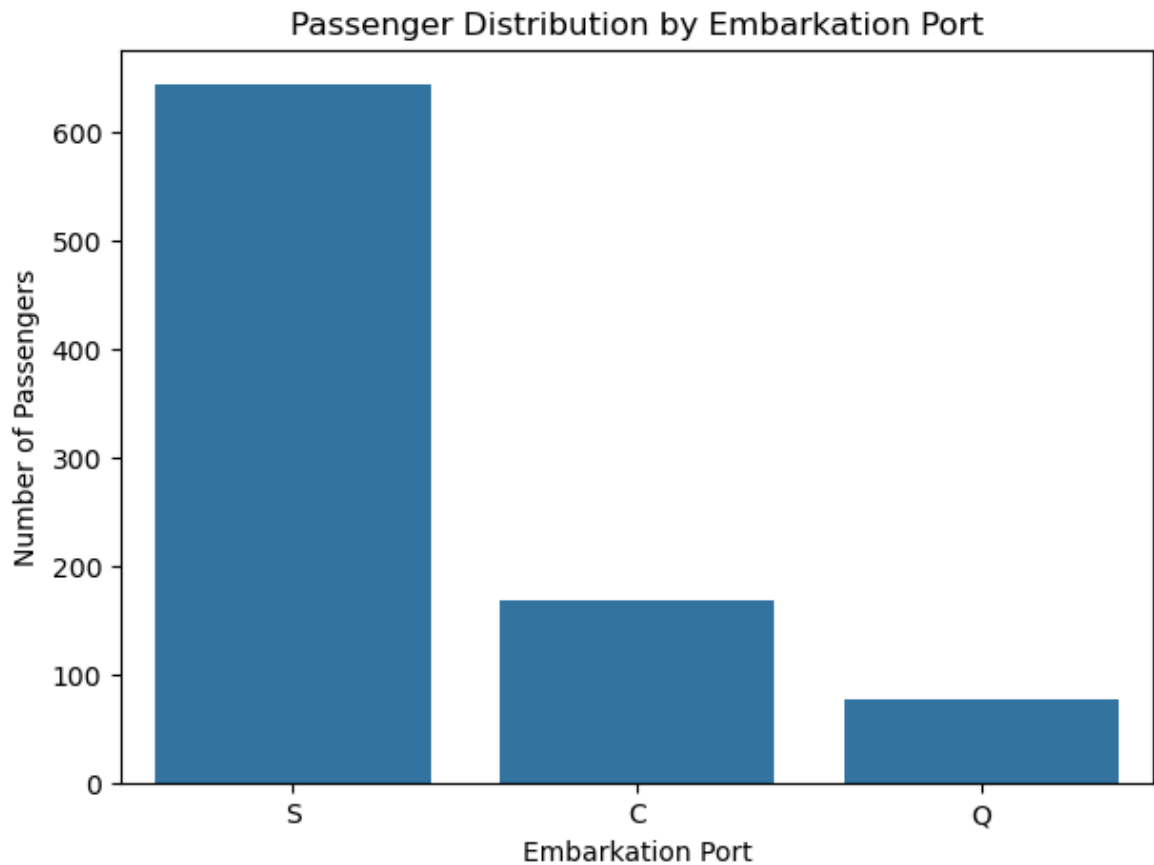
```
In [30]: # Embarkation Port Distribution

plt.figure(figsize=(7, 5))

sns.countplot(data=df, x="embarked")

plt.title("Passenger Distribution by Embarkation Port")
plt.xlabel("Embarkation Port")
plt.ylabel("Number of Passengers")

plt.show()
```



Observation:

The bar chart shows the number of passengers who embarked from each port. It helps identify which embarkation port had the highest and lowest number of passengers.

```
In [31]: # Survival Rate by Gender

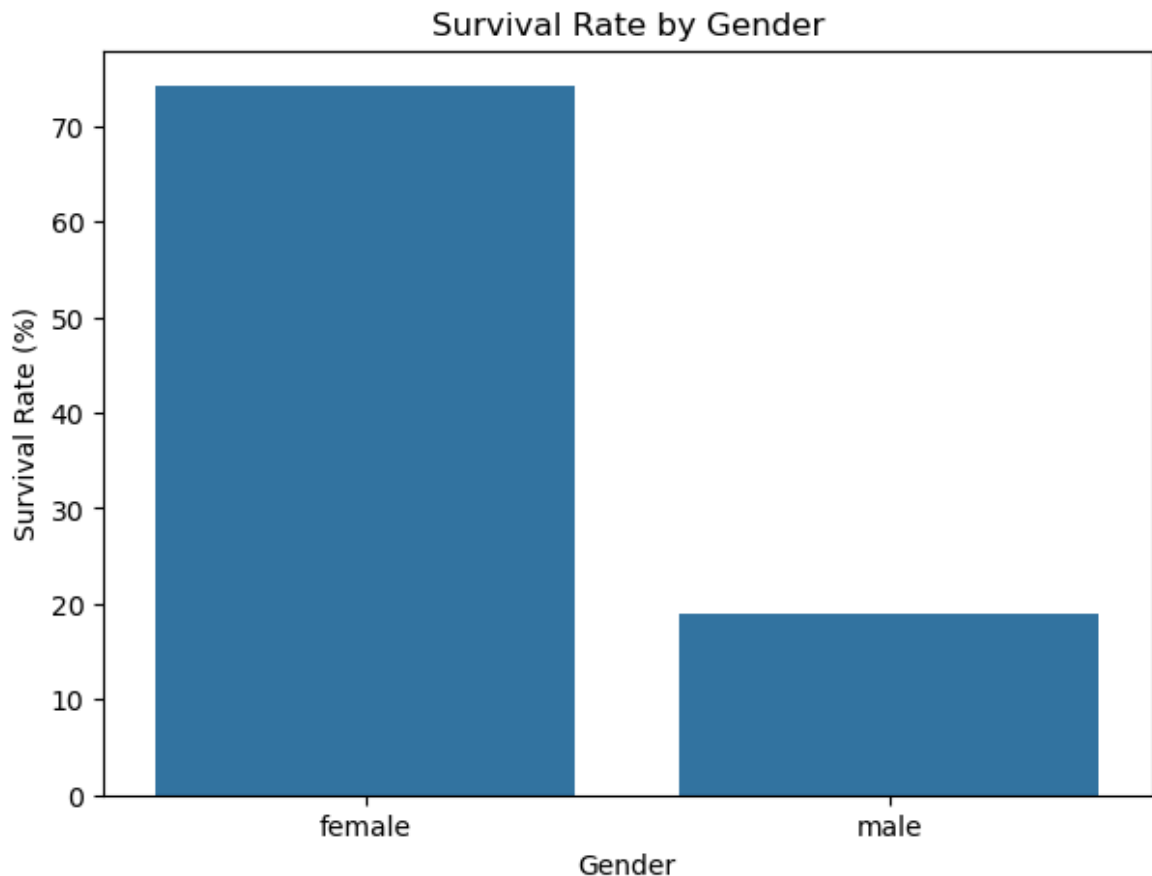
gender_survival_rate = df.groupby("gender")["survived"].mean() * 100

plt.figure(figsize=(7, 5))

sns.barplot(
    x=gender_survival_rate.index,
    y=gender_survival_rate.values
)

plt.title("Survival Rate by Gender")
plt.xlabel("Gender")
plt.ylabel("Survival Rate (%)")

plt.show()
```



Observation:

The bar chart shows the survival rate of passengers by gender. Female passengers had a higher survival rate than male passengers, indicating a strong difference in survival outcomes between the two gender groups.

```
In [32]: # Create age groups

df["age_group"] = pd.cut(
    df["age"],
    bins=[0, 12, 18, 35, 60, 100],
    labels=["Child", "Teenager", "Young Adult", "Adult", "Senior"]
)

# Display age groups
print(df["age_group"].value_counts())
```

```
age_group
Young Adult    358
Adult          195
Teenager        70
Child           69
Senior          22
Name: count, dtype: int64
```

Observation:

The age column has been divided into meaningful age groups: Child, Teenager, Young Adult, Adult, and Senior. This grouping makes it easier to compare survival patterns

across different age categories.

```
In [33]: # Survival Rate by Age Group

age_survival_rate = df.groupby("age_group", observed=True)["survived"].mean() *

print("Survival Rate by Age Group:")
print(age_survival_rate)
```

```
Survival Rate by Age Group:
age_group
Child          57.971014
Teenager       42.857143
Young Adult    38.268156
Adult          40.000000
Senior        22.727273
Name: survived, dtype: float64
```

Observation:

The survival rate varies across different age groups. Comparing these groups helps identify whether age was associated with different survival outcomes among Titanic passengers.

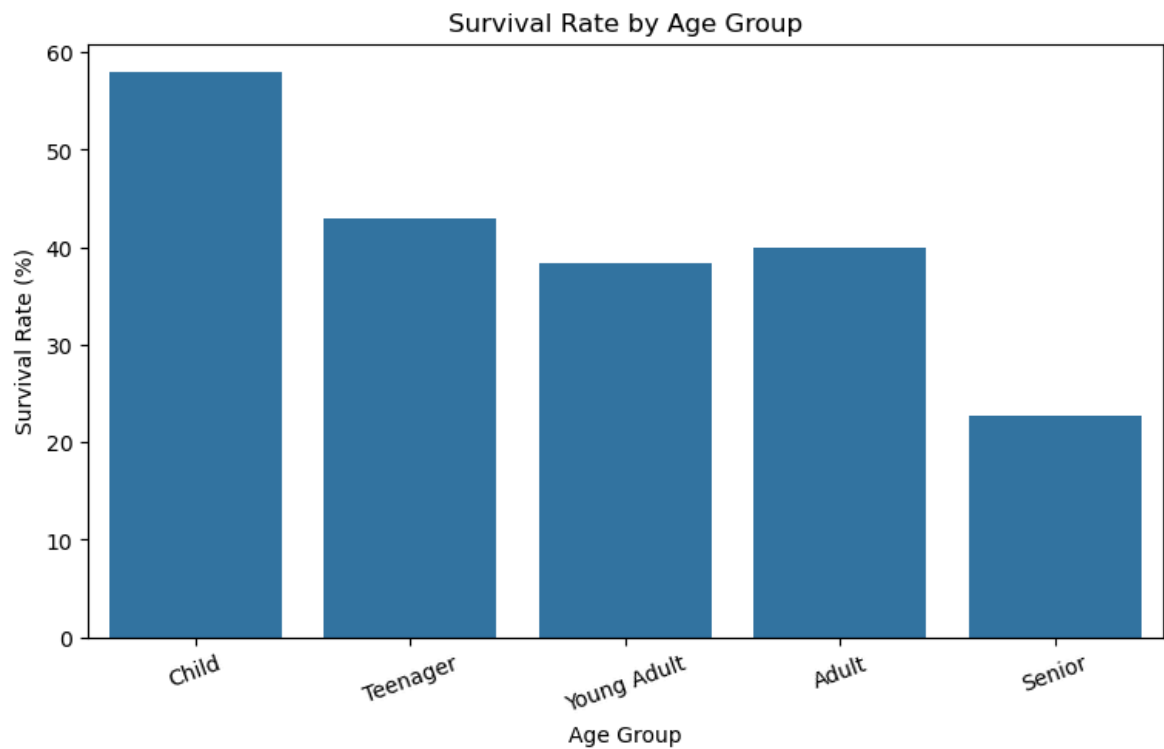
```
In [34]: # Visualize Survival Rate by Age Group

plt.figure(figsize=(9, 5))

sns.barplot(
    x=age_survival_rate.index,
    y=age_survival_rate.values
)

plt.title("Survival Rate by Age Group")
plt.xlabel("Age Group")
plt.ylabel("Survival Rate (%)")
plt.xticks(rotation=20)

plt.show()
```



Observation:

The bar chart compares survival rates across the different age groups. It provides a clear visual comparison of survival outcomes among children, teenagers, young adults, adults, and seniors.

In [35]: *# Multivariate Analysis: Gender, Passenger Class and Survival*

```
plt.figure(figsize=(9, 6))

sns.countplot(
    data=df,
    x="pclass",
    hue="gender",
    palette="Set2"
)

plt.title("Gender Distribution Across Passenger Classes")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")

plt.show()
```

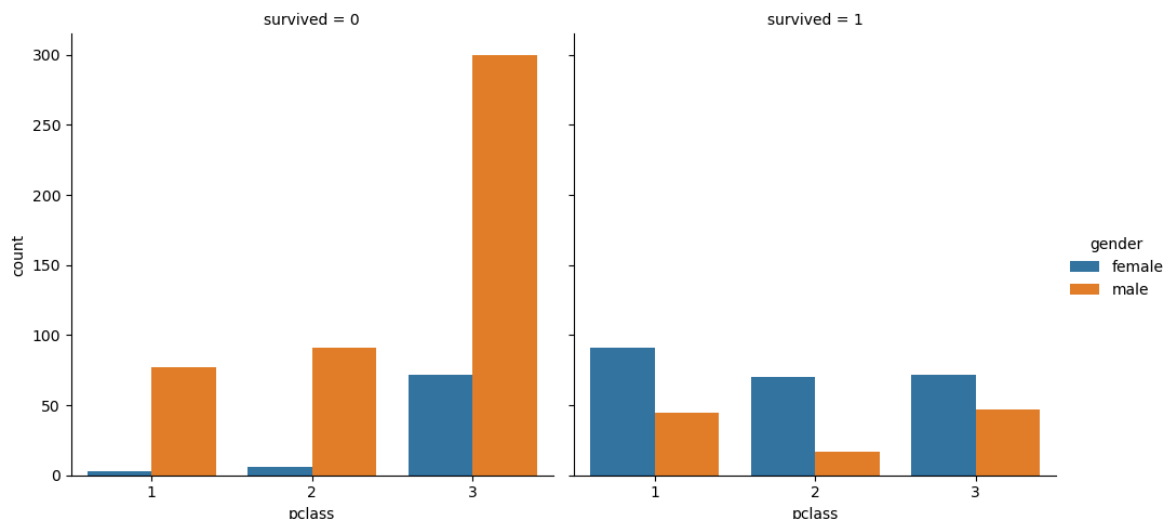


Observation:

The chart compares gender distribution across different passenger classes. It helps identify how male and female passengers were distributed among First, Second, and Third Class.

In [36]: *# Survival by Passenger Class and Gender*

```
sns.catplot(  
    data=df,  
    x="pclass",  
    hue="gender",  
    col="survived",  
    kind="count",  
    height=5  
)  
  
plt.show()
```



Observation:

The multivariate analysis shows how passenger class, gender, and survival are related. It provides a combined view of survival patterns and helps identify differences between male and female passengers across different passenger classes.

Final EDA Findings and Summary

Key Findings

1. The Titanic dataset contains 891 passenger records and 15 columns.
2. Missing values were found mainly in the Age, Embarked, Deck, and Embark_town columns. The Deck column contains the largest number of missing values.
3. The dataset contains both numerical and categorical variables.
4. The number of passengers who did not survive was higher than the number of passengers who survived.
5. Gender had a strong association with survival. Female passengers had a higher survival rate than male passengers.
6. Passenger class was associated with survival. First Class passengers had a higher survival rate, while Third Class passengers had a lower survival rate.
7. The age distribution shows that most passengers belonged to young and middle-age groups.
8. The fare distribution contains some unusually high values, which appear as potential outliers in the boxplot.
9. The correlation heatmap helped identify relationships between numerical variables such as passenger class, age, fare, and survival.

10. The pairplot provided a visual overview of relationships and distributions among multiple numerical variables.
11. Multivariate analysis showed how gender, passenger class, and survival outcomes were related.

Conclusion

Exploratory Data Analysis helped identify important patterns, relationships, distributions, missing values, and possible outliers in the Titanic dataset. The analysis shows that factors such as gender and passenger class were associated with differences in survival outcomes.

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